

Sheet 2 — Vector spaces

Numbering continues from Sheet 1. The former Ex. 15 (logarithms) was removed because it is outside the syllabus.

16. Let $\mathbb{K}$ be a field. On the set $\mathbb{K}^n$, $n \in \mathbb{N}$, define the operations

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \quad \alpha(x_1, \dots, x_n) = (\alpha x_1, \dots, \alpha x_n),$$

for any $x_1, \dots, x_n, y_1, \dots, y_n, \alpha \in \mathbb{K}$. Show that $\mathbb{K}^n$ is a vector space over $\mathbb{K}$.

17. Let P_2 be the set of all real polynomials of degree less than or equal to two. Show that, with the usual operations of addition of polynomials and multiplication of a polynomial by a real number, P_2 is a real vector space.

18. Check whether $\mathbb{R}^3$ is a real vector space when equipped with the following operations:

a) $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2), \quad a(x_1, y_1, z_1) = (ax_1, ay_1, 0);$

b) $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2 + 5, z_1 + z_2), \quad a(x_1, y_1, z_1) = (ax_1, ay_1, az_1).$

19. In the real Cartesian space $\mathbb{R}^3$, consider the vectors $\mathbf{u}_1 = (1, 2, 0)$, $\mathbf{u}_2 = (1, 1, 1)$ and $\mathbf{u}_3 = (0, 0, 1)$. Find the generic vector $\mathbf{v}$ that is a linear combination of $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$, in terms of the coefficients $a, b, c \in \mathbb{R}$.

20. Let $\mathbf{u} = (-1, 1)$, $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, 4)$ be vectors of the real Cartesian space $\mathbb{R}^2$. Show that $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

21. In $\mathbb{R}^3$, consider the vectors $\mathbf{u} = (1, 1, 2)$, $\mathbf{v} = (1, 2, 1)$ and $\mathbf{w} = (3, 4, k)$.

a) Find the value of k for which $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

b) Describe, by means of a condition on x , y and z , the set of vectors (x, y, z) that are linear combinations of $\mathbf{u}$ and $\mathbf{v}$, i.e., describe $L(\{\mathbf{u}, \mathbf{v}\})$.

22. In $\mathbb{R}^3$, consider the sets $A = \{(x, y, z) : x + 2y + z = 0\}$ and $B = \{(x, y, z) : x + 4y - z = 0\}$. Show that $A \cap B$ is a vector subspace of $\mathbb{R}^3$.

23. In the real vector space P_3 of polynomials in x of degree less than or equal to 3, consider the set A of the polynomials of P_3 that vanish at $x = 1$ and $x = -1$. Show that A is a subspace of P_3 .

24. Find the subspaces of $\mathbb{R}^3$ spanned by the following sets of vectors:

a) $A = \{(1, 2, 1)\}$

b) $B = \{(1, 0, 1), (0, 1, 1)\}$

c) $C = \{(1, 0, 1), (0, 1, 1), (1, 1, 2)\}$

d) $D = \{(1, 0, 1), (0, 1, 1), (1, 0, 0)\}$

25. Find the subspace of P_2 spanned by the vectors $p_1(x) = x + x^2$ and $p_2(x) = 2 + x$.

26. In the real vector space $\mathcal{F}(\mathbb{R})$ of all real functions with domain $\mathbb{R}$, let A be the subspace spanned by the vectors $f(x) = e^x$ and $g(x) = e^x + 1$. Check whether the vector $h(x) = e^{x+1} + 3e^x + 2$ belongs to A .

27. Study the linear dependence of the following vectors of $\mathbb{R}^2$: $\mathbf{u} = (1, 2)$, $\mathbf{v} = (0, 1)$ and $\mathbf{w} = (1, 4)$.

28. Check whether the following sets of vectors are linearly independent:

a) $\{(1, 2, 3), (4, 5, 6)\}$

b) $\{(2, -10, 6), (-1, 5, -3)\}$

c) $\{(0, 1, 3), (1, 2, 1), (1, 1, 2)\}$

d) $\{(0, 1, 3), (1, 2, 1), (1, 1, 2), (1, 1, 1)\}$

29. Show that the following elements of P_2 are linearly independent: $p_1(x) = 1 + x$, $p_2(x) = 2 - x^2$, $p_3(x) = x - x^2$.

30. Check whether the following vectors of $\mathbb{R}^3$ are linearly independent (solve using the theorems):

a) $\mathbf{u} = (1, 1, 1)$, $\mathbf{v} = (0, 0, 1)$ and $\mathbf{w} = (1, 1, 8)$

b) $(0, 1, 3)$, $(0, 0, 0)$, $(1, 1, 2)$

c) $\mathbf{u} = (1, 1, 1)$, $\mathbf{v} = (-3, -3, -3)$ and $\mathbf{w} = (1, 0, 1)$

31. In a real vector space U , the vectors $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$ are known to be linearly independent. Study the linear dependence of the following sets of vectors:

- a) $\{\mathbf{u}_1, \mathbf{u}_2\}$ b) $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{0}\}$ c) $\{\mathbf{u}_1, -3\mathbf{u}_2, \mathbf{u}_3\}$
 d) $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_1 + \mathbf{u}_2 + 7\mathbf{u}_3\}$ e) $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_1 + \mathbf{u}_2\}$

32. Prove that $\{(1, 1), (0, 1)\}$ is a basis of $\mathbb{R}^2$.

33. In $\mathbb{R}^3$, consider the vector subspace $A = \{(x, y, z) \in \mathbb{R}^3 : 2x - y + z = 0\}$.

- a) Show that $\{(1, 2, 0), (0, 1, 1), (1, 3, 1)\}$ is a spanning set of A , but is not a basis of A .
 b) Find a basis and the dimension of A .
 c) Show that the vector $\mathbf{u} = (2, 5, 1)$ belongs to A and give its coordinates in the basis found in b), after ordering it.

34. Show that $\{2, 1 + x, x + x^2\}$ is a basis of P_2 : a) assuming the dimension of P_2 is unknown; b) using the fact that the dimension of P_2 is 3.

35. Find a basis for each of the following vector spaces, and give its dimension:

- a) $A = \{(x, y, z) \in \mathbb{R}^3 : 2x + y - 3z = 0\}$
 b) $B = \{(a - 2c, b + c - 2a, 2c - a, a + c - b) : a, b, c \in \mathbb{R}\}$

36. Consider the subspace of $\mathbb{R}^4$: $A = \{(x, y, z, w) \in \mathbb{R}^4 : 2x + y = 0 \wedge x + y - w = 0\}$.

- a) Find the dimension of A .
 b) Show that $((1, -2, 1, -1), (2, -4, 0, -2))$ is a basis of A and find the coordinates of the generic element of A in that basis.

37. Consider the subspace of $\mathbb{R}^4$: $A = \{(x, y, z, w) \in \mathbb{R}^4 : 3x + y = 0 \wedge 2x + 3y + w = 0\}$.

- a) Find the dimension of A .
 b) Show that $((2, -6, 0, 14), (1, -3, 1, 7))$ is a basis of A and find the coordinates of the generic element of A in that basis.

38. In P_2 , consider the basis $B = (2, 1 + x, x + x^2)$.

- a) Find the coordinates of the vector $p(x) = x^2 + 3x + 8$ in the basis B .
 b) Consider the vector $q(x)$ whose coordinates relative to the basis B are $(1, 2, 1)$. Find the coordinates of $q(x)$ in the basis $C = (x, -2, x^2)$.
 c) What are the coordinates of $(x - 1)^2$ in the basis $D = (1 + x, 2 - x^2, x - x^2)$? And in the basis $E = (2 - x^2, x - x^2, 1 + x)$?